## SYSTEM PROJECT: Student Record Management System (documentation)

### ER Diagram

| | students | |
|---|---|---|
| PK | id | INT AUTO_INCREMENT |
| UQ | student_number | VARCHAR(20) NOT NULL |
| | first_name | VARCHAR(100) NOT NULL |
| | last_name | VARCHAR(100) NOT NULL |
| | course | VARCHAR(100) NOT NULL |
| | year_level | INT NOT NULL |
| | created_at | TIMESTAMP DEFAULT CURRENT_TIMESTAMP |

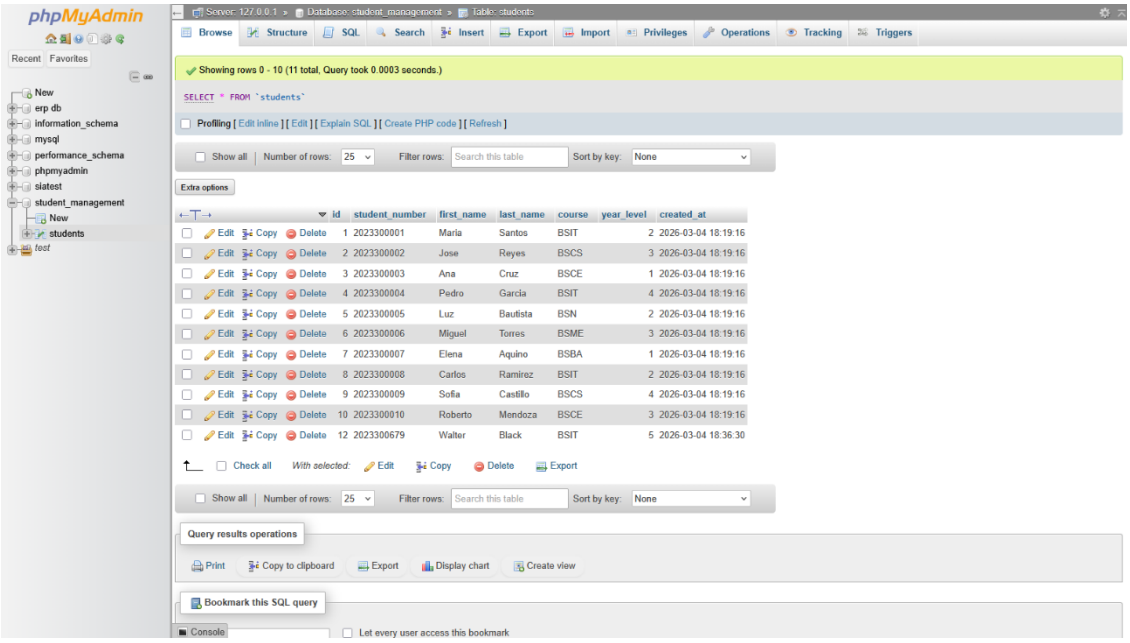
Legend:
PK = Primary Key (Unique identifier for each record)
UQ = Unique Constraint (No duplicates allowed)

### API Documentation

| Method | Endpoint | Description | Status Codes |
|---|---|---|---|
| POST | /api/students | Add new student | 201, 400, 500 |
| GET | /api/students | Get all students | 200, 500 |
| GET | /api/students/:id | Get single student | 200, 404, 409, 500 |
| GET | /api/students/search/:key word | Search by name/ID | 200, 500 |
| PUT | /api/students/:id | Update student | 200, 400, 404, 500 |
| DELETE | /api/students/:id | Delete student | 200, 404, 409, 500 |

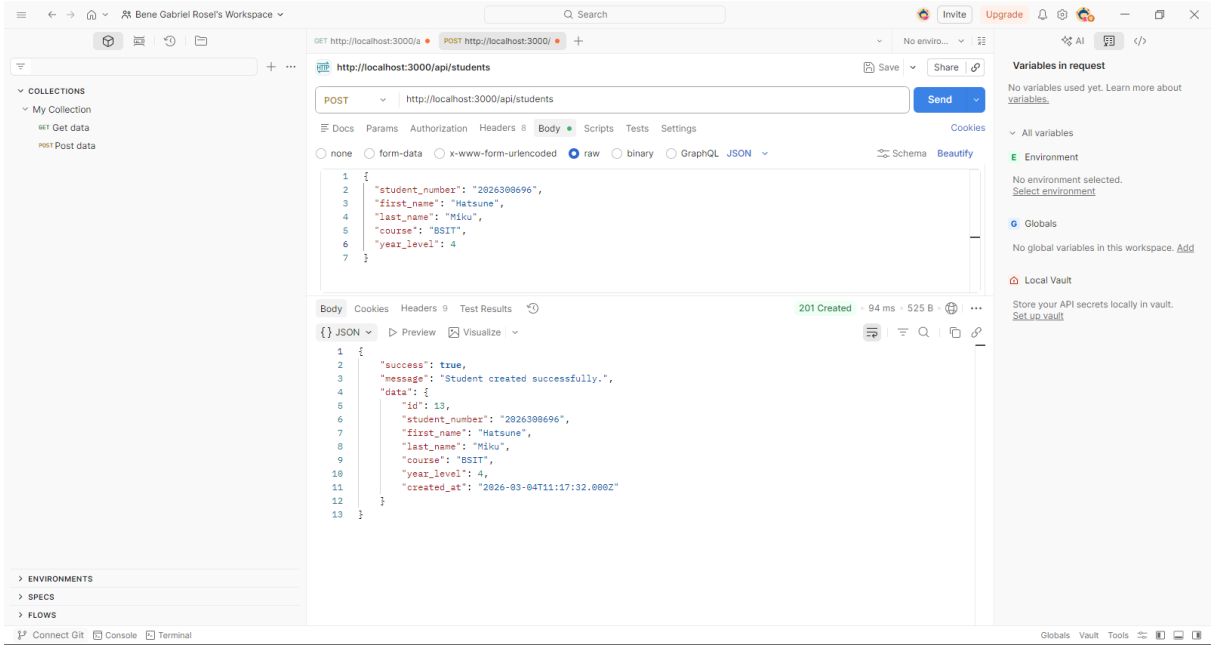
## Screenshots

- Database Table

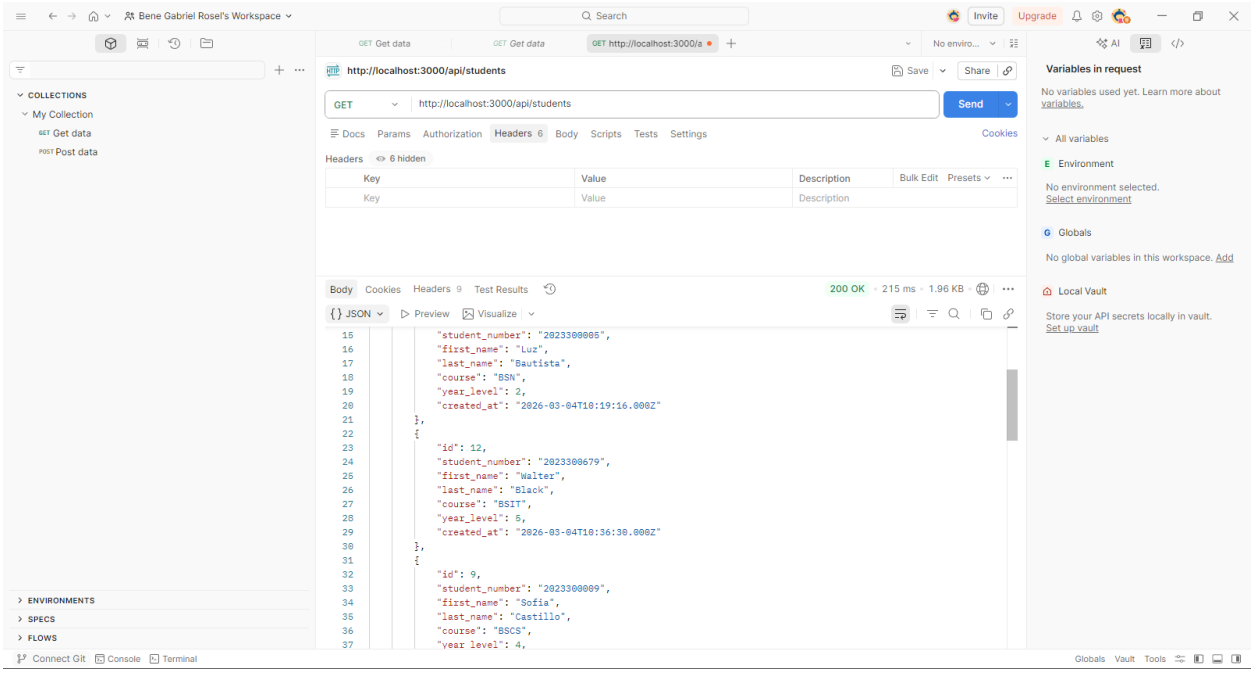



- Postman testing

I. Insert Student (POST)

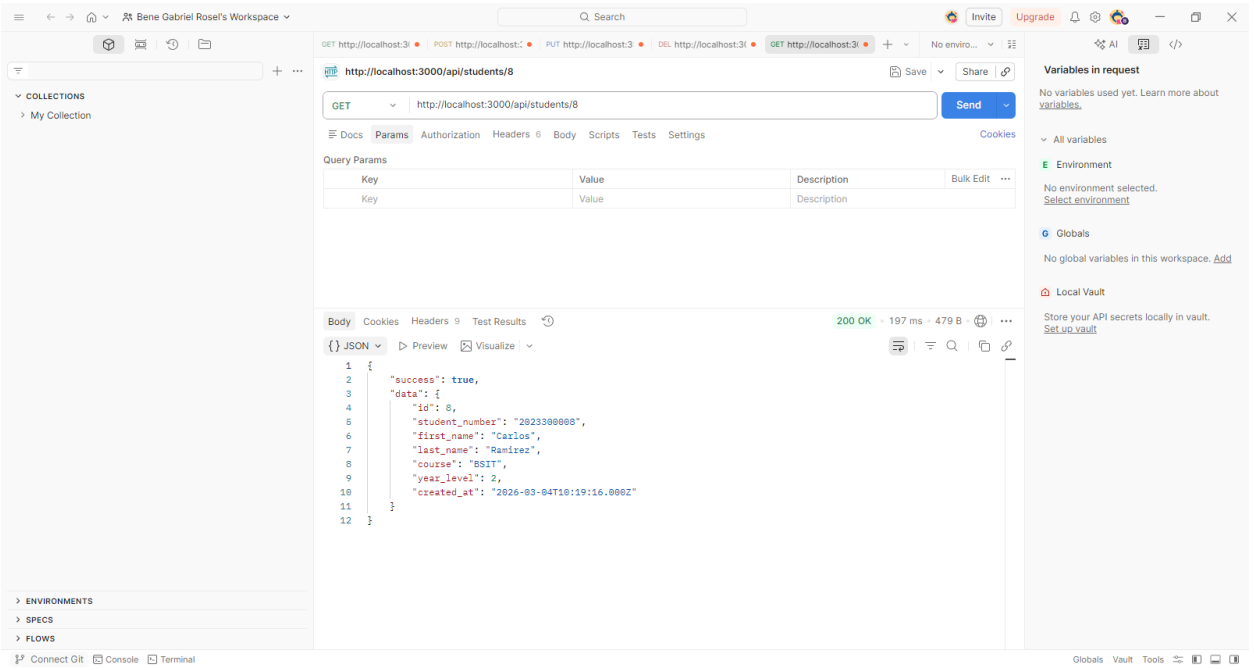


## II. Select all students (GET)

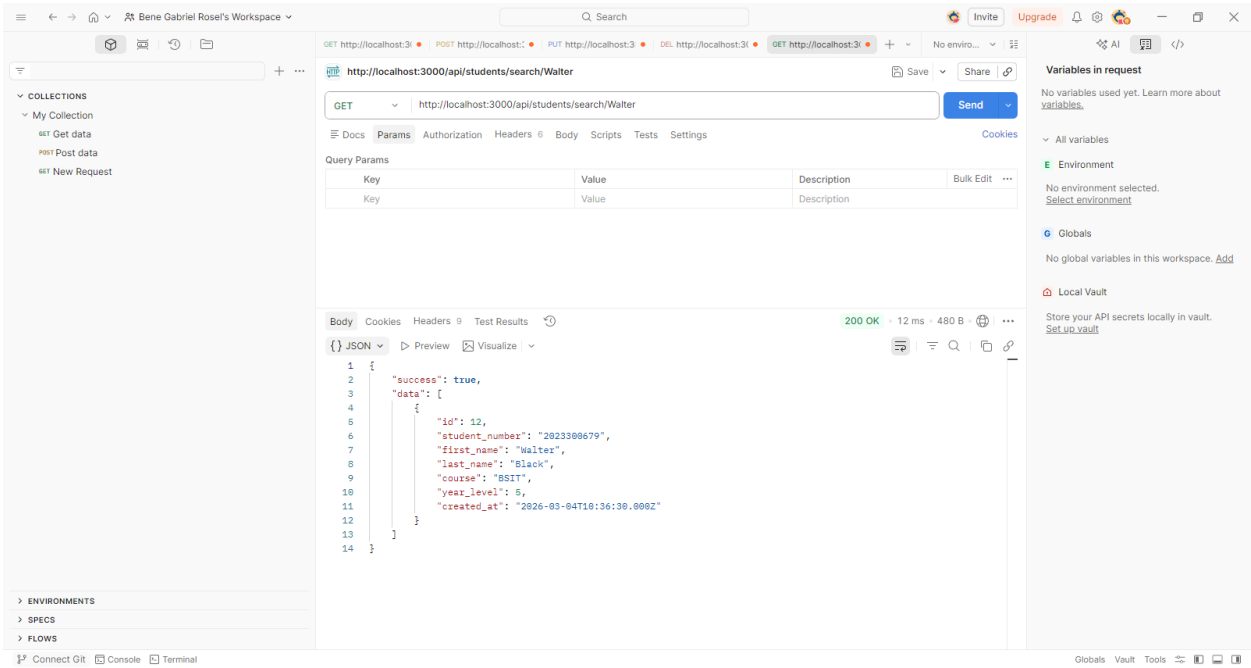



## III. Select student (GET)

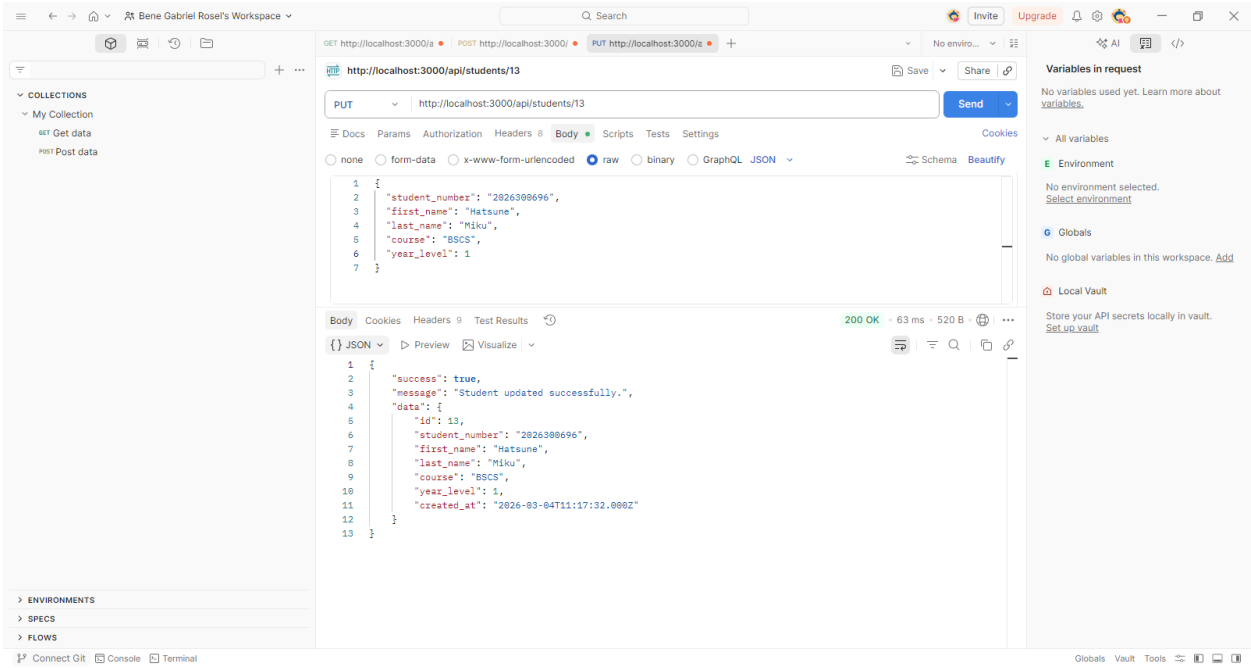
- By ID



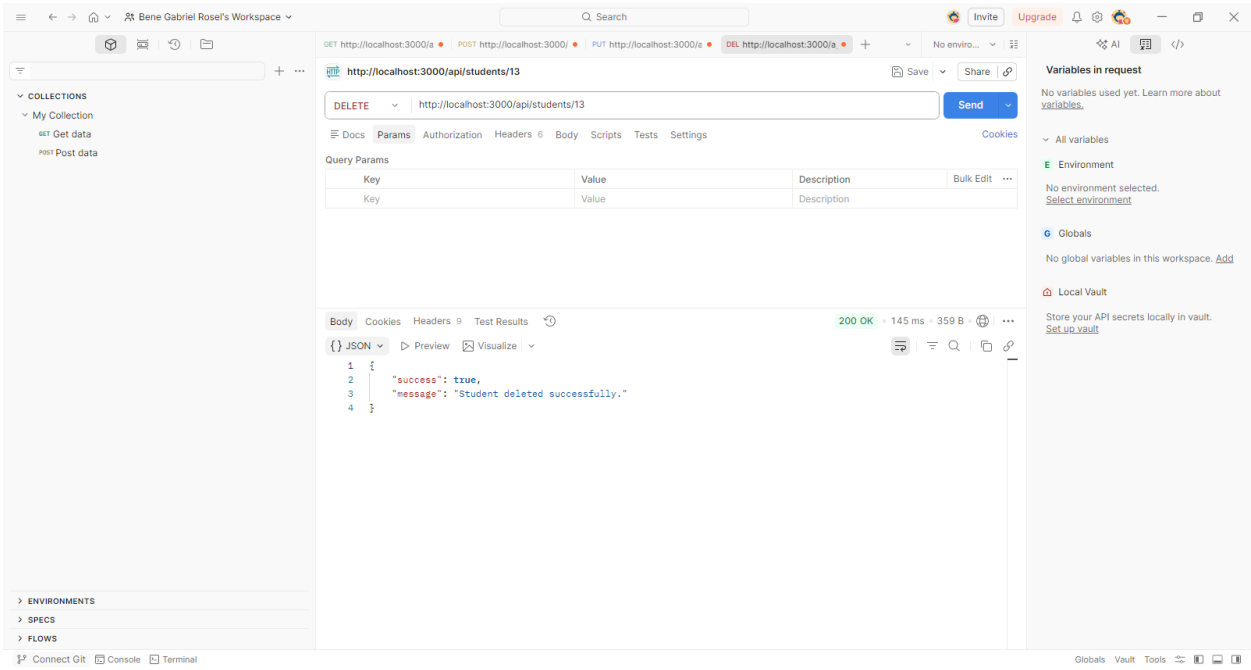
- By name/ID




## IV. Update student (PUT)

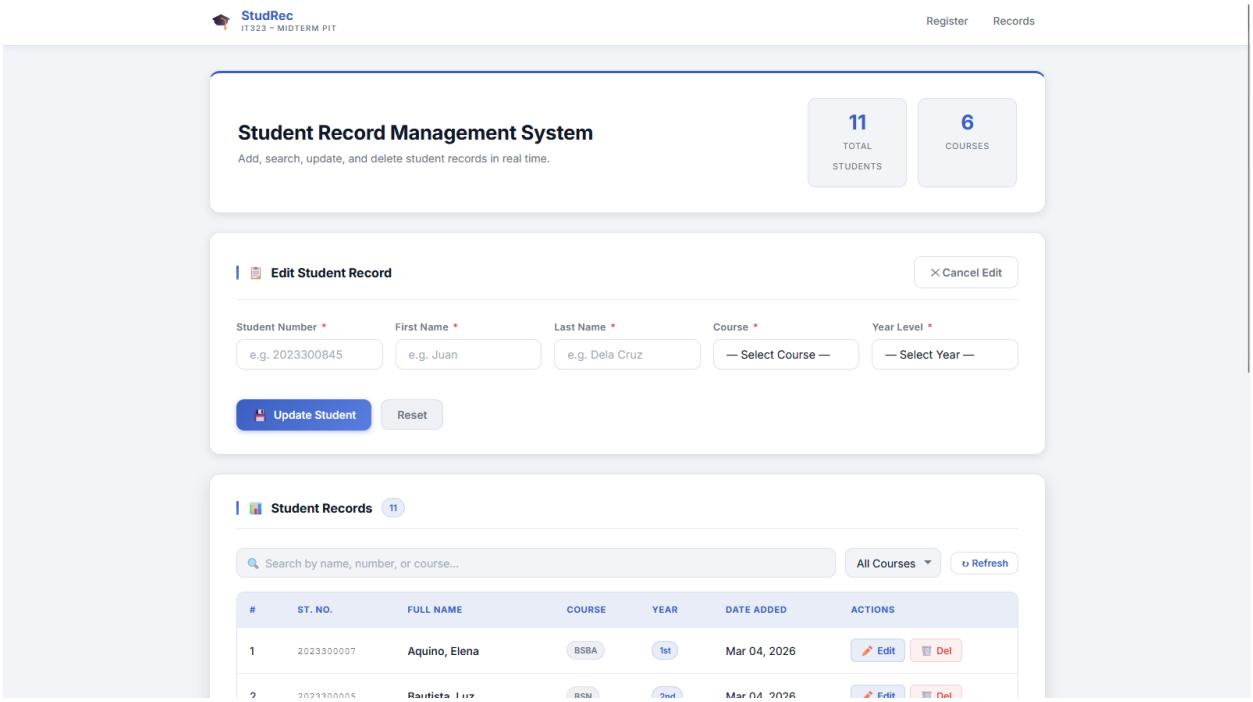


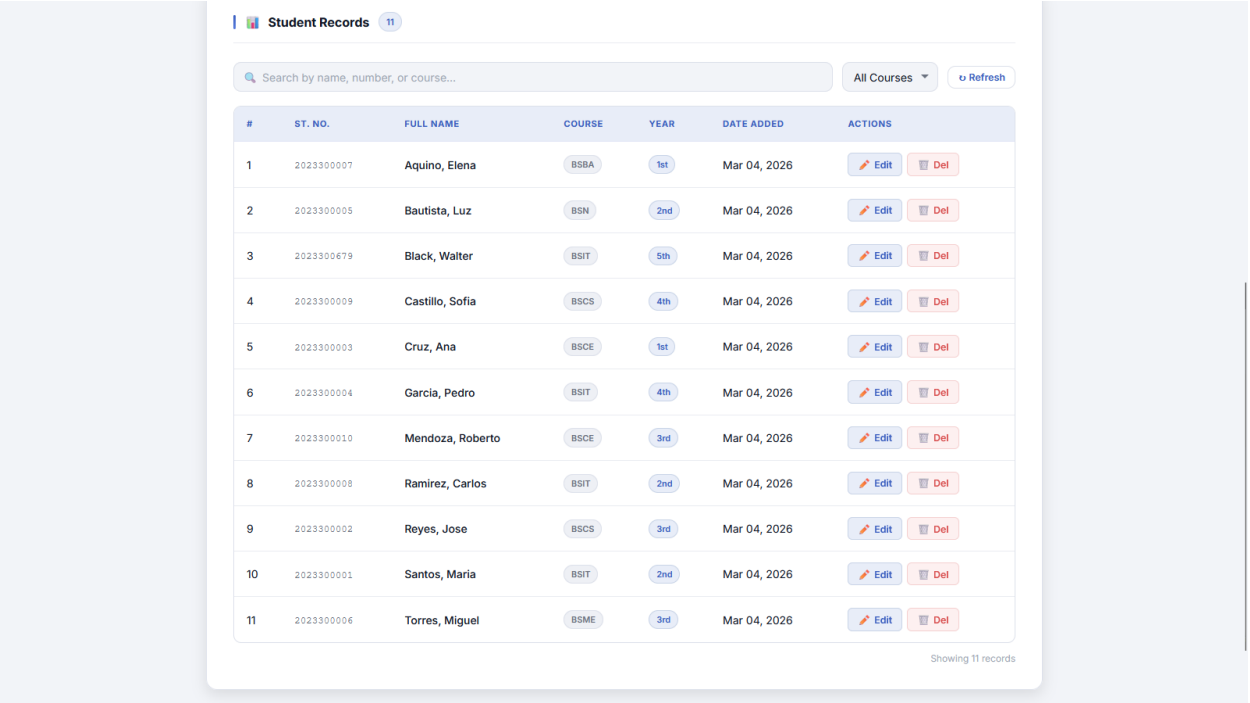
## V. Delete student (DELETE)



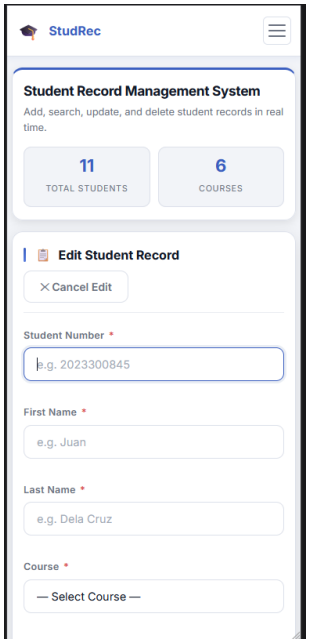

- Responsive Desktop View

## I. Main Pages




Student Records 11
Search by name, number, or course...
All Courses
Refresh
# ST. NO. FULL NAME COURSE YEAR DATE ADDED ACTIONS
1 2023300007 Aquino, Elena BSBA 1st Mar 04, 2026 Edit Del
2 2023300005 Bautista, Luz BSN 2nd Mar 04, 2026 Edit Del
3 2023300679 Black, Walter BSIT 5th Mar 04, 2026 Edit Del
4 2023300009 Castillo, Sofia BSCS 4th Mar 04, 2026 Edit Del
5 2023300003 Cruz, Ana BSCE 1st Mar 04, 2026 Edit Del
6 2023300004 Garcia, Pedro BSIT 4th Mar 04, 2026 Edit Del
7 2023300010 Mendoza, Roberto BSCE 3rd Mar 04, 2026 Edit Del
8 2023300008 Ramirez, Carlos BSIT 2nd Mar 04, 2026 Edit Del
9 2023300002 Reyes, Jose BSCS 3rd Mar 04, 2026 Edit Del
10 2023300001 Santos, Maria BSIT 2nd Mar 04, 2026 Edit Del
11 2023300006 Torres, Miguel BSME 3rd Mar 04, 2026 Edit Del
Showing 11 records

- Mobile View

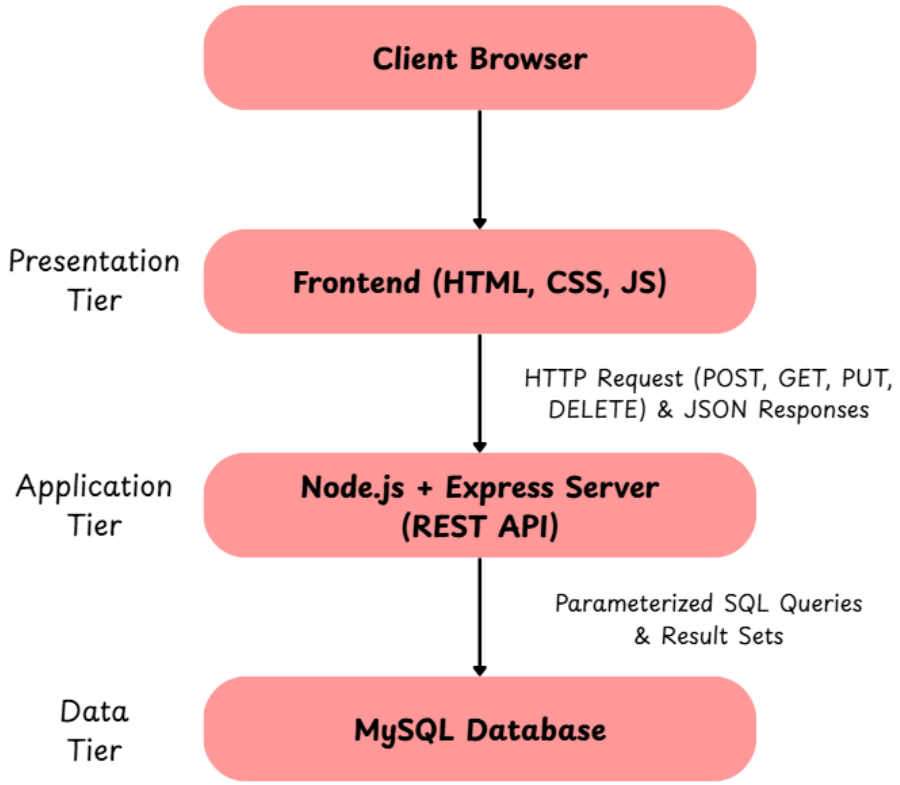

StudRec
Student Record Management System
Add, search, update, and delete student records in real time.
11
TOTAL STUDENTS
6
COURSES
Edit Student Record
Cancel Edit
Student Number *
e.g. 2023300845
First Name *
e.g. Juan
Last Name *
e.g. Dela Cruz
Course *
— Select Course —

ST. NO. FULL NAME COURSE ACTIONS
2023300007 Aquino, Elena BSBA Edit Del
2023300005 Bautista, Luz BSN Edit Del
2023300679 Black, Walter BSIT Edit Del
2023300009 Castillo, Sofia BSCS Edit Del
2023300003 Cruz, Ana BSCE Edit Del
2023300004 Garcia, Pedro BSIT Edit Del
2023300010 Mendoza, Roberto BSCE Edit Del
2023300008 Ramirez, Carlos BSIT Edit Del
2023300002 Reyes, Jose BSCS Edit Del
2023300001 Santos, Maria BSIT Edit Del
2023300006 Torres, Miguel BSME Edit

## Technical Report

**• System Architecture**

The Student Record Management System (SRMS) is a full-stack web application built following **the three-tier architecture** model




Project Structure

- server.js - Entry point, mounts middleware + routes
- package.json - Dependencies and npm scripts
- .env.example - Environment variable template (copy to .env)
- config/db.js - MySQL connection pool (mysql2/promise + dotenv)
- routes/students.js - All /api/students route handlers
- public/
  index.html - Single-page frontend
  style.css - Full design system (Grid + Flex)
  script.js - All Vanilla JS logic
- student_management.sql - Database schema + seed data

- **Database Design Rationale**

I. Schema Design

| Column | Data Type | Constraints | Description |
|---|---|---|---|
| id | INT | AUTO_INCREMENT PRIMARY KEY | Unique row identifier |
| student_number | VARCHAR(20) | UNIQUE NOT NULL | Official student ID number |
| first_name | VARCHAR(100) | NOT NULL | Student's first name |
| last_name | VARCHAR(100) | NOT NULL | Student's last name |
| course | VARCHAR(100) | NOT NULL | Enrolled Academic Program |
| year_level | INT | NOT NULL CHECK (year_level BETWEEN 1 AND 5), | Current year level (1-4) |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Record creation timestamp |

- The AUTO_INCREMENT primary key ensures each record has a unique, system-generated identifier that is safe to use in API routes.
- The UNIQUE constraint on student_number enforces business rules at the database level, preventing duplicate enrollment entries even if application- level validation is bypassed.

II. SQL Operation implemented

| Operation | SQL statement used |
|---|---|
| Insert student | INSERT INTO students (student_number, ...) VALUES (?, ?, ...) |
| Select all student (ordered ASC) | SELECT * FROM students ORDER BY last_name ASC |
| Select student by ID | SELECT * FROM students WHERE id = ? |
| Update student | UPDATE students SET student_number=?, first_name=?, last_name=?, course=?, year_level=? WHERE id=? |
| Delete student | DELETE FROM students WHERE id = ? |
| Search (using LIKE) | SELECT * FROM students WHERE first_name LIKE ? OR last_name LIKE ? OR student_number LIKE ? OR course LIKE ? |

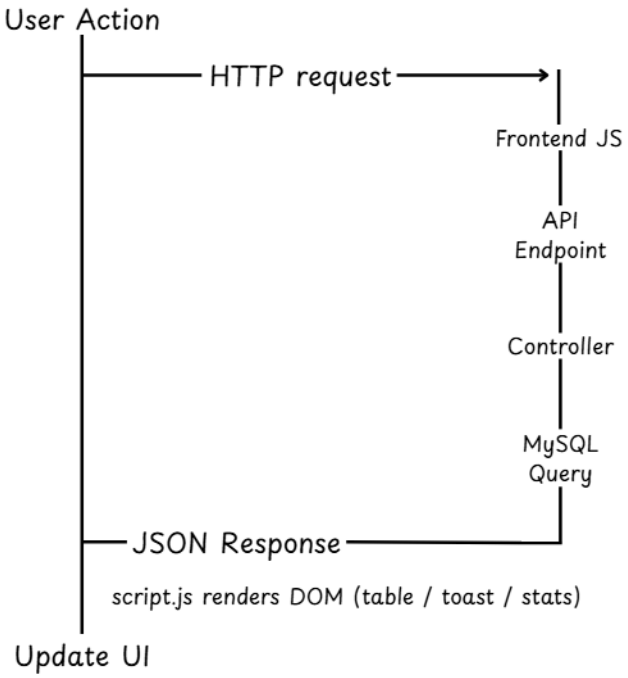
III. Normalization

1NF: All columns are atomic; no repeating groups

2NF: No partial dependencies (single-column PK)

3NF: No transitive dependencies; `course` is a free-text attribute intentionally not normalized into a separate table at this stage

- **API Flow Explanation**




  - Error Handling Strategy
    400 - Missing required fields or duplicate `student_number` (caught via `ER_DUP_ENTRY` MySQL error code)
    404 - Row lookup before update/delete; returns early with message
    500 - Unexpected DB errors; caught in `try/catch`, logged server-side, generic message returned to client

- **Challenges Encountered**

  - Problem: Express matched `/api/students/search/:keyword` as `/api/students/:id` (interpreting `"search"` as an ID), triggering a MySQL type mismatch error.
    Solution: The `search/:keyword` route was declared **above** the `/:id` route in `routes/students.js`. Express evaluates routes in registration order, so placing the specific literal path first ensures it matches before the generic wildcard.
  - Problem: Firing an API call on every keystroke caused excessive server requests and visual flickering.
    Solution: Implemented a 400 ms debounce timer in `script.js` using `clearTimeout` / `setTimeout`. The search API is only called once the user pauses typing, reducing network calls by ~80% during normal use.
  - Problem: Rendering student data directly via `innerHTML` with unsanitized data could allow stored XSS if malicious content was inserted into the database.

Solution**:** An `escHtml()` utility function was implemented to escape `&`, `<`, `>`, `"`, and `'` characters before any user-provided data is inserted into the DOM via `innerHTML`.

- Problem: At viewport widths ≤768px, the navigation links overlapped with the brand and badge elements, breaking the layout.
  Solution: A hamburger button (`#nav-toggle`) was added to the navbar. On mobile, the nav links are hidden by default and revealed as a full-width vertical dropdown when the hamburger is clicked. The button animates into a close (✕) icon when the menu is open using CSS-only bar transforms.

**• Future Improvements**

- User Authentication
  Implement JWT-based authentication with role-based access control (admin vs. registrar) using `jsonwebtoken` and `bcrypt` for password hashing.
- Pagination
  Add server-side pagination (`LIMIT / OFFSET`) for scalability when the student population grows beyond a few hundred records. The API would accept `?page=1&limit=20` query parameters.
- Normalized Courses Table
  Create a separate `courses` master table with a foreign key from `students.course_id`, enabling course management (add/archive courses) without modifying student records directly
- Data Export
  Allow administrators to export the student list as a CSV or PDF report via a `GET /api/students/export` endpoint using `json2csv` or `pdfmake`.
- Automated Testing
  Add unit tests for route handlers using `Jest` + `supertest`, and integration tests against a dedicated test database to ensure regression safety during development.

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